

Performance Testing

Date	02 October 2026
Team ID	SWTID-2026-1380
Project Name	PocketSmartAI - Your smart budget and recommendations assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how PocketSmart AI behaves under expected and peak load conditions. This document records the planned performance-testing approach for the FastAPI backend and identifies the measurements to capture during execution.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter (performance/load testing)
Type of Testing	Load Testing
Target Module / API	FastAPI APIs: /api/generate-home and /api/shopping
Test Environment	Local development environment (FastAPI + Uvicorn)
Test Date	02 October 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Home Planner recommendation request under normal load	5	30	Successful responses with acceptable response time and no server errors.
2	Shopping/Product Discovery request with category, budget and platform filters	10	60	Filtered products are returned correctly without request failures.
3	Mixed planner/API requests under increased load	20	60	Application remains responsive and error rate stays below the target.
4	Peak-load API test across recommendation endpoints	50	60	Server handles peak requests without crashes or significant degradation.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.558 seconds	Pass	JMeter Summary Report: average response time 558 ms.
2	Response Time (Max)	< 5 seconds	1.003 seconds	Pass	JMeter Summary Report: maximum response time 1003 ms.
3	Throughput (Req/sec)		2.0 req/sec	Measured	JMeter Summary Report: throughput 2.0 requests/sec.
4	Error Rate	< 1%	0%	Pass	JMeter Summary Report: 0.00% errors across 20 samples.
5	CPU Utilization	< 80%	Not measured	Not Tested	CPU utilization was not captured in the supplied JMeter screenshot.
6	Memory Utilization	< 80%	Not measured	Not Tested	Memory utilization was not captured in the supplied JMeter screenshot.

Step 4: Observations & Analysis

Current status: A JMeter Summary Report screenshot was captured for the PocketSmartAI home recommendation API run. The report shows 20 samples, an average response time of 558 ms, a maximum of 1003 ms, 0.00% errors, and throughput of 2.0 requests/sec. CPU and memory utilization were not captured in the supplied evidence.

Key Findings: The captured run completed 20 samples with 0.00% errors, 558 ms average response time, 1003 ms maximum response time, and 2.0 requests/sec throughput.

Bottlenecks Identified: No request failures are visible in the supplied Summary Report. CPU and memory utilization were not captured.

Optimization Steps Taken: No optimization change was recorded from the supplied evidence; additional load runs can be used for further tuning.

Step 5: Screenshots / Evidence

JMeter evidence from the executed PocketSmartAI performance run is attached below.

The screenshot shows the JMeter Summary Report window. The left sidebar displays the test plan structure, including a Thread Group with a Recording Controller and several HTTP requests. The main panel shows the Summary Report for the selected thread group. The report includes a table with columns for Label, # Samples, Average, Min, Max, Std. Dev., Error %, Throughput, Received KB/sec, Sent KB/sec, and Avg. Bytes. The data shows a total of 20 samples with an average response time of 558 ms and a maximum of 1003 ms. The error rate is 0.00%.

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
/api/generate-h...	20	558	366	1003	139.20	0.00%	2.0/sec	0.80	1.56	409.0
TOTAL	20	558	366	1003	139.20	0.00%	2.0/sec	0.80	1.56	409.0

Figure: JMeter Summary Report — PocketSmartAI